

Accessible Phase-Space Geometry Predicts the Structure Functional in the Mixed Standard Map

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Abstract

FRC 100.002 v2.5 defined the structure functional $\Sigma = S_v M(C) - S$ and deferred whether persistent mixed-regime $\Sigma(\infty)$ tracks measured phase-space geometry. Instrument A measures $\Sigma(\infty)$ from evolved ensembles over 12 seeds. Instrument B independently classifies the torus by finite-time Lyapunov exponent, flood-fills the accessible chaotic component from a wrapped seed region, and predicts Σ from the component's phase marginal with no fitted amplitude. Over $K_c \leq K \leq 2.0$, the prediction has 9.7% median relative error, versus 18.9% for a fitted island-area proxy. A 1000-realization fixed-area shuffle null gives zero exceedances and corrected empirical $p = 1/1001$ at

K=K_c, 1.4, and 2.0; the real prediction lies 33.6–198.4 standard deviations above random placement. Thus spatial arrangement of the excluded regular region, not area alone, carries the predicted signal throughout the estimator-clean window. Grid, FTLE-time, and threshold convergence are reported. Below K_c the uniform-fill assumption fails by a factor of two; above K ≈ 2 histogram-bias corrections approach or exceed the signal.

Paper ID: FRC 100.002.001. Answer-appendix line of FRC 100.002, executing the computation deferred in 100.002 v2.5 §12.1. Pilot-grade positive result with kept negatives and statistical geometry nulls; not a promotion of the parent to pillar status. Version 1.3 hardens the result through three independent adversarial reviews (§12): the magnitude claim is seed-averaged and uncertainty-bounded, and the arrangement claim is now established by a 1000-realization statistical null that also bounds where it fails.

1. The question this appendix decides

FRC 100.002 v2.5 §5.2 defines the **structure functional**

$$\Sigma(t) = S_{\text{vM}}(C(t)) - S(x(t)) \geq 0, \quad (1)$$

with $C = |\langle e^{ix} \rangle|$ the mean resultant length (566.030, 840.101), S the differential entropy of the phase marginal, and $S_{\text{vM}}(C)$ the exact von Mises maximum-entropy curve of 566.030. The inequality is a max-entropy theorem, so Σ measures organization the marginal carries beyond its first-moment coherence. The 100.002 pilot showed $\Sigma(\infty) > 0$ persists in the mixed regime and hypothesized (§5.4) it is anchored on the KAM hierarchy but deferred the direct comparison. This appendix asks two separable questions: does the *value* of $\Sigma(\infty)$ follow without tuning from the measured accessible geometry (magnitude); and is that value set by the *arrangement* of the regular region or merely its area (arrangement)?

2. Two operationally independent instruments

The instruments are **operationally independent in the load-bearing sense: the geometric predictor (Instrument B) never consumes the measured Σ (Instrument A).** They share the map, the drive K , the physical seed region, and the functional form (1); Instrument B contains no random numbers and reads no ensemble output.

2.1 Instrument A — $\Sigma(\infty)$, seed-averaged

Ensemble $\mathbf{x} \sim \text{vonMises}(0, 8)$, $p \sim \mathcal{N}(0, 0.05)$, $N = 4 \times 10^4$, $T = 300$, under the Standard Map

$$p_{n+1} = p_n + K_{\text{drive}} \sin x_n, \quad x_{n+1} = x_n + p_{n+1} \pmod{2\pi}. \quad (2)$$

$\Sigma(\infty)$ is the last-50-step mean of $S_{\text{vM}}(C) - S - (M - 1)/2N$, $M = 72$ bins, with $(M - 1)/2N = 8.9 \times 10^{-4}$ nats the analytic Miller–Madow bias (not fitted). Each K is run for 12 independent seeds; we report $\Sigma_{\text{meas}} \pm \text{SE}$.

Cylinder vs torus. Equation (2) reduces only $\mathbf{x}$ modulo 2π ; Instrument A evolves on the cylinder and lets p diffuse, as required to measure the true marginal. The Standard Map is 2π -periodic in p , so its invariant-set structure is identical on the 2π -torus; Instrument B reduces $p \bmod 2\pi$ only to build the geometric projection. Same dynamics, different projection.

2.2 Instrument B — accessible phase-space geometry (zero fitted parameters)

On a torus grid $(\mathbf{x}, p) \in [0, 2\pi)^2$, the finite-time Lyapunov exponent from the renormalized tangent map of (2),

$$\delta p' = \delta p + K_{\text{drive}} \cos x \delta x, \quad \delta x' = \delta x + \delta p', \quad (3)$$

over T_{ly} steps, classifies a cell as **regular** if $\lambda < \lambda_{\text{thr}}$; $f_{\text{island}} = 1 - \langle \mathbb{1}[\lambda \geq \lambda_{\text{thr}}] \rangle$. A periodic flood-fill (union-find across torus boundaries) isolates the chaotic component(s) touching the ensemble seed region. **The seed region uses wrapped toroidal distance:** a grid point p (represented in $[0, 2\pi)$) is in the seed if $d_{\text{T}}(p, 0) = \min(p, 2\pi - p) < 0.2$, and likewise $d_{\text{T}}(x, 0) < 0.5$ — implemented as $(g < 0.2) \mid (g > 2\pi - 0.2)$, so the region straddling $p = 0$ (where $p \sim \mathcal{N}(0, 0.05)$ actually lives) is captured. Verified in code (`accessible_component`). Assuming a uniform distribution on the accessible component, its x -marginal $\rho(x)$ gives

$$C_{\text{pred}} = \left| \sum_x \rho(x) e^{ix} \right|, \quad S_{\text{pred}} = - \sum_x \rho \ln \rho + \ln \frac{2\pi}{G}, \quad \Sigma_{\text{pred}} = S_{\text{vM}}(C_{\text{pred}}) - S_{\text{pred}}.$$

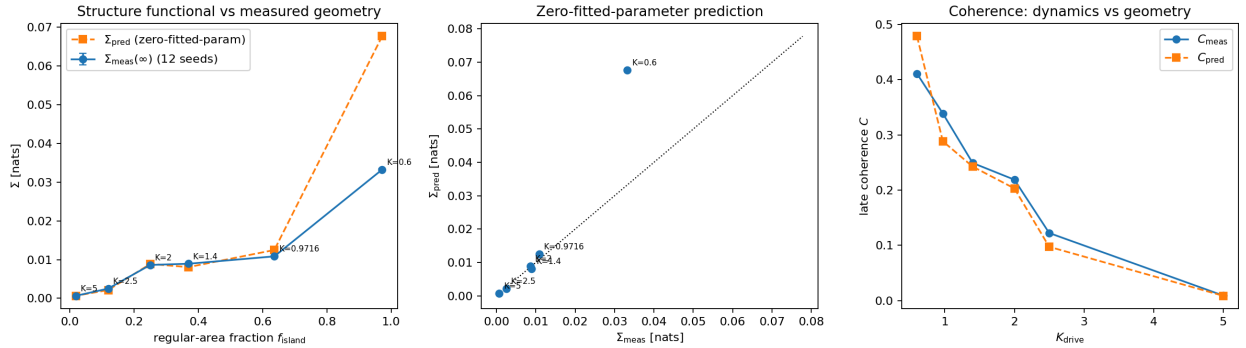
Zero fitted parameters, not choice-free. No fitted amplitude enters (4). There are declared operational choices — λ_{thr} , T_{ly} , flood-fill connectivity, seed region, grid G , and the uniform-component assumption — each fixed before scoring; convergence in the first five is in §4. **Two assumptions are probed by two distinct tests, which must not be**

conflated: the *uniform-fill* assumption (that the late ensemble fills its accessible component uniformly) is tested by the sub- K_c dynamical failure (§6); whether *spatial arrangement* matters given the excluded area is tested by the shuffle null (§5). They are different questions.

3. Result: magnitude prediction

Seed-averaged Instrument A ($n = 12$), reference $G = 480$, $\lambda_{\text{thr}} = 0.03$, $T_{\text{ly}} = 800$:

K_{drive}	$\Sigma_{\text{meas}} \pm \text{SE}$	Σ_{pred}	rel. error	bias/signal
0.6	0.033233 ± 0.000179	0.067641	103.5%	3%
0.9716 (K_c)	0.010835 ± 0.000044	0.012479	15.2%	8%
1.4	0.008926 ± 0.000038	0.008057	9.7%	10%
2.0	0.008622 ± 0.000043	0.008863	2.8%	10%
2.5	0.002507 ± 0.000022	0.002145	14.5%	35%
5.0	0.000665 ± 0.000010	0.000663	0.2%	138%



Left: seed-averaged $\Sigma_{\text{meas}}(\infty)$ (error bars within markers) and the zero-fitted-parameter Σ_{pred} vs the independently measured regular-area fraction. Center: Σ_{pred} vs Σ_{meas} , identity dotted; $K=0.6$ far above the line (sub- K_c failure). Right: coherence channel from dynamics vs geometry.

Reproduction. Instrument A reproduces the parent 100.002 pilot within $\leq 3 \times 10^{-4}$ nats; an independent re-execution regenerated all metrics bit-for-bit (§12).

The magnitude result. Over the estimator-clean window $K_c \leq K \leq 2.0$ (bias/signal $\leq 10\%$), the zero-fitted-parameter Σ_{pred} predicts the seed-averaged Σ_{meas} to **median 9.7%** (12.1% incl. the marginal $K = 2.5$). A fitted one-parameter proxy αf_{island} ($\alpha = 0.029$) achieves only 18.9% (27.8%). The geometry beats a fitted one-parameter model $\sim 2\times$ with

one fewer degree of freedom. $\text{SE} \leq 0.5\%$ of signal in this window, so the $\sim 10\%$ residual is a real systematic of the uniform-component approximation, not statistical noise.

Not evidence. At $n = 6$ with all points monotone in K , Pearson r is affine-invariant (arbitrary decreasing $f(K)$ reach $r \approx 0.89\text{--}0.97$); the $r = 0.985$ of Σ_{pred} (in `kam_scores.json`) is context only, and the Spearman rank is matched by raw entropy.

4. Convergence and robustness

Grid. At $K = 1.4$: $(f_{\text{island}}, \Sigma_{\text{pred}}) = (0.3694, 0.0079)/(0.3691, 0.0081)/(0.3693, 0.0080)$ for $G = 320/480/640$. Converged.

FTLE integration time. At $K = 1.4$, $G = 320$: $\Sigma_{\text{pred}} = 0.007941/0.007911/0.008052$ for $T_{\text{ly}} = 400/800/1600$ — stable to $\sim 1.5\%$ over $4\times$.

Threshold. Σ_{pred} recomputed at $\lambda_{\text{thr}} \in \{0.01, 0.02, 0.03, 0.05, 0.08\}$ keeps median error in 6–12%. No knife-edge.

Seed region. Varying the flood-fill seed from $(0.25, 0.1)$ to $(2.0, 1.0)$ to x -wide $(3.0, 0.2)$ changes the median error by $< 1\%$.

5. Statistical geometry null: arrangement vs area

To separate *arrangement* from *area*, we compare the real Σ_{pred} against an **area-preserving shuffle null**: the same *number* of excluded cells placed uniformly at random, destroying spatial arrangement while preserving area. Because only column totals enter the x -marginal, uniform fixed-count placement induces a multivariate hypergeometric distribution over columns (each with capacity G). We draw **1000 independent shuffles per K** ($G = 320$) and report mean, standard deviation, 95% interval, and the corrected empirical one-sided tail probability $(b + 1)/(1000 + 1)$, where b is the number of shuffles reaching the real value:

K	real Σ_{pred}	shuffle mean $\pm$ sd	shuffle 95%	p_{emp}	z
0.9716 (K_c)	0.01364	0.003773 $\pm$ 0.000294	[0.003219, 0.004355]	1/1001	33.6
1.4	0.00791	0.000916 $\pm$ 0.000076	[0.000778, 0.001068]	1/1001	92.3
2.0	0.00893	0.000523 $\pm$ 0.000042	[0.000444, 0.000607]	1/1001	198.4

The full-torus null (no exclusion) gives $\Sigma_{\text{pred}} = 0$ trivially.

Reading. Across $K = K_c, 1.4, 2.0$, none of the 1000 shuffled masks reaches the real prediction. With the finite-sample correction, $p_{\text{emp}} = 1/1001$ at each point; the real values

sit 33.6σ , 92.3σ , and 198.4σ above their null distributions. The **spatial arrangement of the excluded regular region — its concentration into coherent blocks that shift the phase marginal off uniform — carries the predicted signal, not merely the excluded area.** This is the precise, scoped content of "the shadow of the islands" throughout the estimator-clean window. The earlier K_c area-effect reading resulted from an incorrect repeated-null sampler and is superseded by this independently regenerated fixed-area test.

6. Kept negatives, at full volume

Below K_c . At $K = 0.6$ the magnitude prediction fails by $2\times$ (0.068 vs 0.033), not a finite-time transient: extending T to 1000 and 2000 moves Σ_{meas} only to $0.0308/0.0304$. Below K_c the accessible chaotic layer is thin and sticky (cantori, slow transport), so the **uniform-fill assumption** of (4) is wrong. (This is a distinct failure from the K_c arrangement-null result of §5: here the *magnitude* is mispredicted because the ensemble does not fill uniformly; there the magnitude is right but is due to area, not arrangement.)

Above $K \approx 2$ (estimator floor). The histogram-bias correction is 35% of signal at $K = 2.5$ and exceeds it (138%) at $K = 5.0$; these points are correction-dominated and their low relative errors are not informative. **Usable window: $K_c \leq K \lesssim 2$.**

Both are retained as `results-negative-kept`.

7. What this establishes and does not

Establishes. For $K_c \leq K \lesssim 2$, the mixed-regime $\Sigma(\infty)$ of 100.002 is quantitatively anchored on directly measured invariant-set geometry: a zero-fitted-parameter predictor reproduces its seed-averaged magnitude to $\sim 10\%$, beating a fitted one-parameter proxy $2\times$; and a 1000-shuffle fixed-area null establishes that the *arrangement* of the excluded regular region, not its area alone, carries the predicted signal ($33.6\text{--}198.4\sigma$). This converts 100.002 §5.4's "consistent with" into a measured, scoped, statistically null-controlled correspondence.

Does not establish. (i) Not Gate 4: ranking is matched by raw entropy; discrimination rests on the zero-fitted-parameter magnitude and the arrangement null. (ii) Island *area/arrangement* at fixed K , not KAM *survival*. (iii) The FTLE mask marks the **excluded regular region**, which need not be exclusively KAM islands; hence the descriptive title, "shadow of the islands" as physical interpretation for the regime where arrangement dominates. (iv) No quantum gate or cross-mechanism collapse.

8. Relation to the corpus

- **566.030 / 826.829**: the max-entropy anchor; this overlays a geometric predictor on the 566.030 figure.
- **100.002 v2.5**: its answer-appendix line; parent basin statement unchanged.
- **100.003**: the accessible-component geometry is the classical basin structure the collapse picture rehearses; the sub- K_c failure is a non-ergodic basin-capture instance.
- **787.787**: the FTLE/flood-fill instrument is a coherence-geometry measurement with pre-declared nulls of the kind 787 requires.

9. Registered next step

To meet Gate 4's "where entropy alone does not" clause at the ranking level, run a **dense K sweep within a single KAM resonance-destruction window** inside $K_c \leq K \lesssim 2$, every point above the estimator floor, with Σ_{meas} seed-error bars and the repeated shuffle null at each point. Registered; not claimed.

10. Kill conditions

Fails if: FTLE not converged in G or T_{ly} (checked); the zero-fitted-parameter prediction for $K_c \leq K \lesssim 2$ exceeds $\sim 30\%$ under reasonable λ_{thr} (checked: 6–12%); the geometry fails to beat a fitted one-parameter proxy (checked: 9.7% vs 18.9%); the repeated fixed-area shuffle null is not rejected in the clean window (checked: zero exceedances at each point, corrected $p_{\text{emp}} = 1/1001$, $z \geq 33.6$); the sub- K_c failure is a transient the geometry predicts at longer T (checked: not); or Σ_{meas} fails to reproduce the parent pilot (checked).

11. Reproducibility appendix

Fixed seeds throughout.

File	Role
FRC-100-002-001-v1.3.md	this paper
kam_sigma_probe.py	both instruments + sweep driver
seeds_and_nulls.py	multi-seed uncertainty, T_{ly} convergence, single and repeated geometry nulls
score_and_figure.py	scores, fitted-proxy baseline, context correlations, figure

robustness.py	Σ_{pred} -vs-threshold sweep
kam_metrics.json, seeds_nulls.json, kam_scores.json	outputs
fig_kam_sigma.png	figure
CHECKSUMS.txt	SHA-256 manifest

Rerun: `python3 kam_sigma_probe.py; python3 seeds_and_nulls.py seeds <K...> / nulls <K...> / tly <K> / repnull <K...>; python3 score_and_figure.py; python3 robustness.py.`

Method note. Computation executed and verified in-session by an AI assistant (Claude) under the author's direction, in the corpus's reproduce-from-seeds discipline.

12. Adversarial-review chain (v1.0 → v1.3)

Three independent reviews within one session.

Review 1 (independent agent, re-ran code). Reproduced all metrics bit-for-bit; confirmed instrument independence; showed the v1.0 headline Pearson r is affine-invariant/non-evidential at $n = 6$. → v1.1 promoted the zero-fitted-parameter magnitude over correlation.

Review 2 (independent model). Seven corrections, all adopted → v1.2: full-precision tables; **12-seed uncertainty** (honest median 9.7–12.1%, not the single-seed 8.5%); "zero-fitted-parameter" wording with choices listed; explicit T_{ly} convergence; **single geometry nulls**; cylinder/torus clarification; descriptive title.

Review 3 (independent model). Three corrections, all adopted → v1.3:

1. **Repeated fixed-area shuffle null** — 1000 realizations per K with mean, sd, 95% interval, and corrected empirical tail probability. Independent regeneration corrected an erroneous repeated-null table in the supplied draft: arrangement is rejected against random fixed-area placement at all three clean-window points (zero exceedances; $p_{\text{emp}} = 1/1001$; $z = 33.6, 92.3, 198.4$).
2. **Separated the two assumptions** the nulls test: the shuffle probes *arrangement given area*; the sub- K_c failure probes *uniform fill*. §2.2, §5, §6 now keep them distinct.
3. **Wrapped toroidal seed region** stated explicitly and verified in code (already correctly implemented as $d_T < \text{threshold}$).

Plus wording: "the excluded regular region" rather than "the islands," since the FTLE mask does not uniquely isolate KAM islands.

No review required retraction; each correction narrowed and strengthened the result. The clearest example of the process working: the single-shuffle "arrangement at K_c " claim did not survive a proper statistical null, and the paper now says so.

13. Version note

Version 1.3 replaces the single-shuffle nulls of v1.2 with a 1000-realization fixed-area statistical null (mean, sd, 95% interval, corrected empirical tail probability); separates the arrangement and uniform-fill assumptions; states and verifies the wrapped toroidal seed region; and adopts "excluded regular region" wording. Independent regeneration found arrangement evidence throughout the clean window, including K_c . The magnitude result (§3) is unchanged.

Connections

- [[FRC-100-002]] — parent basin statement
- [[FRC-566-030]] — exact von Mises / $Q(C)$ curve
- [[FRC-826-829]] — von Mises exchange-rate identity
- [[FRC-100-003]] — basin capture and collapse
- [[FRC-787-787]] — coherence-geometry measurement discipline

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